SESSION 4 — MORNING

Corporate Finance

Corporate structure • Capital allocation • WACC • Capital structure theory

Session Roadmap — 2 Hours

Corporate finance bridges FSA and equity valuation. It explains how firms are structured, how they invest, and how they finance those investments.

01 **Corporate Structure**
Org forms • public vs private • going public (IPO) • going private

02 **Stakeholders & ESG**
Debt vs equity claims • agency conflicts • environmental, social and governance factors (CFA LM2)

03 **Corporate Governance**
Board roles • conflicts of interest • mechanisms • executive compensation

04 **Capital Allocation (CFA LM5)**
NPV: Gerhardt €13.136m • IRR: 19.52% vs 6.76% • multiple IRRs • ROIC: 8.73%

05 **WACC & Capital Costs**
WACC formula • Nutry 8.2% (CFA LM6) • leverage comparison Firm A vs B

06 **Capital Structure Theory**
MM propositions • Plover Corp V_L=€107m WACC=7.48% • trade-off • pecking order

Learning Objectives

- **Describe Organisational Forms**
  Compare sole proprietor, partnership, limited company. Explain key features of corporate issuers (legal identity, owner-manager separation, limited liability).
- **Stakeholders, Governance & ESG**
  Distinguish debt vs equity claims. Identify principal-agent conflicts. Describe ESG factors and how they affect investment decisions.
- **Apply Capital Allocation Tools**
  Calculate NPV and IRR. Resolve NPV-IRR conflicts. Recognise multiple IRRs. Compute ROIC and compare to required return (all CFA LM5 examples).
- **Calculate WACC**
  Apply the WACC formula. Solve Nutry Company (8.2%) from CFA LM6. Understand how leverage affects cost of equity.
- **Apply Leverage Analysis**
  Compare Firm A (20% debt, ROE=35%) vs Firm B (60% debt, ROE=53%) under different OI scenarios. Explain financial leverage risk-return trade-off.
- **Explain Capital Structure Theories**
  State MM Propositions I/II with and without taxes. Solve Plover Corp (V_L=€107m, WACC=7.48%). Contrast static trade-off with pecking order theory.

SECTION 01

Corporate Structure (CFA LM1)

Organisational Forms of Business — CFA LM1

The choice of organisational form determines liability, taxation, access to capital and governance

Form	Description	Key Advantage	Key Limitation
Sole Proprietorship	One owner, no legal separation from business	Simple, full control, no corporate tax	Unlimited liability, hard to raise capital, dies with owner
General Partnership	Two or more owners, share management	Flexible profit sharing, pass-through tax	Unlimited joint liability, dissolution on partner exit
Limited Partnership	General + limited partners (passive investors)	Limited partners have limited liability	General partner still unlimited liability
Limited Liability Company (LLC)	Limited liability for all members, pass-through tax	Best of corporation + partnership	More complex to establish than partnership
Corporation (Limited Company)	Separate legal entity, shares issued to owners	Limited liability, perpetual, raises capital easily	Double taxation (corp + personal), governance costs
Public Company	Corporation with shares traded on public exchange	Maximum capital access, liquidity for owners	Regulatory burden, disclosure requirements, agency costs

Public vs Private Companies — CFA LM1

CFA LM1 examines the distinction between publicly traded and privately held corporate issuers. Each model involves a different set of trade-offs.

Public Company Advantages

Access to large amounts of capital through equity issuance and public debt markets.
Shares have market price — liquidity for shareholders and for M&A (stock deals).
Higher profile, brand awareness, recruitment advantage.
Owner diversification: founders can sell partial stake while retaining control.

Private Company Advantages

No disclosure requirements: no quarterly reporting, no proxy statements.
No shareholder activism or short-term market pressure.
Fewer governance costs (audit, board, compliance).
Long-term focus: management can invest without quarterly earnings pressure.
Private equity can provide expertise alongside capital.

Going Public (IPO) vs Going Private

IPO: company files with regulator (SEC/FCA), issues new shares to public via underwriter bookbuild. Price set via demand discovery.
Going private: buyout firm acquires all shares (often with leverage — LBO). Public shareholders exit. Management gains flexibility away from markets.
CFA LM1: SPAC (Special Purpose Acquisition Company) is an alternative route to public markets: a blank-cheque shell company raises capital via IPO to later acquire a private target.

Key Features of Corporate Issuers (CFA LM1)

Five features distinguish corporations from other business forms. Each has direct implications for investors and creditors.

Legal Identity

A corporation is a legal person distinct from its owners. It can own property, sue/be sued, enter contracts, and continue beyond the death of any shareholder. Unlike a sole proprietorship, the business does not end when an owner dies.

Owner-Manager Separation

Shareholders own the firm but hire managers to run it. This creates AGENCY PROBLEM: managers' interests may not align with shareholders'. Remedied by incentive alignment (stock options, performance pay), monitoring (board, auditors), and market discipline (takeover threat).

Limited Liability

Shareholders' maximum loss = their investment. Creditors cannot pursue personal assets of shareholders. This enables risk-taking and broad equity ownership. Critical feature for attracting passive investors.

External Financing

Corporations can raise capital from a broad range of external investors via equity (IPO, SEO) and debt (bonds, loans). This is the fundamental advantage of incorporation over sole proprietorship.

Taxation

Corporations face entity-level taxation on profits (corporation tax). Dividends paid to shareholders are often taxed again at personal level (double taxation). This creates incentive to distribute via buybacks instead of dividends.

SECTION 02

Stakeholders & ESG (CFA LM2)

Financial Claims — Debt vs Equity (CFA LM2)

Debt = fixed priority claim on cash flows. Equity = residual claim after all obligations. Higher risk = higher return.

Feature	Debt Investors	Equity Investors
Claim type	Fixed, contractual. Interest + principal repayment.	Residual. Only receive what remains after all debts paid.
Priority in liquidation	Priority claim on assets. Secured > unsecured.	Receive value only AFTER all creditors are paid.
Return profile	Capped: maximum = promised interest. No upside.	Unlimited upside if company thrives. Full downside loss.
Risk level	Lower: contract limits loss to non-recovery	Higher: full loss of investment in bankruptcy possible
Governance rights	Covenants: restrictions on leverage, coverage, etc.	Voting rights on board elections and major transactions
Maturity	Finite: defined repayment schedule	Perpetual: no fixed maturity date for equity

CFA LM2: 'Equity is a residual claim of indeterminate length'. Debt investors have lower required return because they take less risk — hence why WACC includes after-tax cost of debt.

Agency Conflicts — Managers vs Shareholders (CFA LM2)

Agency problem: managers (agents) act on behalf of shareholders (principals) but may have different interests. CFA LM2 presents multiple conflict types.

Conflict Type	Description	Example	Mitigation
Empire building	Managers expand assets beyond efficient size to increase power/comp	Acquisitions at excessive premium destroy shareholder value	Performance-based pay tied to ROIC not size
Entrenching	Managers make firm-specific investments that only they can manage	R&D in technology CEO understands but no-one else does	Transparent strategy, board oversight
Risk aversion	Managers too conservative to protect their jobs	Pass on high-NPV projects due to career risk if project fails	Equity ownership aligns risk preferences
Excessive perks	Managers extract value via perquisites	Private jet, lavish offices, excessive comp	Audit, compensation committee oversight
Shareholders vs creditors	Shareholders encourage excessive risk at creditors' expense	Issue debt, pay out cash dividend, leaving creditors with riskier firm	Debt covenants limiting distributions/extra debt

ESG Factors — Environmental, Social & Governance (CFA LM2)

CFA LM2 presents ESG as material factors affecting long-term corporate value. Not just ethics — ESG affects cash flows, cost of capital and risk.

E

Environmental

- Climate change: carbon emissions, physical risk, transition risk
- Resource depletion: water, land, biodiversity
- Pollution: air, water, soil contamination
- Supply chain: scope 3 emissions, sourcing practices
- Regulatory: carbon pricing, environmental permits

S

Social

- Labour practices: health & safety, human rights
- Employee relations: diversity, turnover, training
- Supply chain: labour standards in Tier 2/3 suppliers
- Product safety and quality
- Community relations and social licence to operate

G

Governance

- Board composition: independence, diversity, expertise
- Executive compensation: alignment with long-term value
- Shareholder rights: voting, equal treatment
- Transparency and disclosure quality
- Anti-corruption and business ethics policies

SECTION 03

Corporate Governance (CFA LM3)

Corporate Governance Framework — Board of Directors (CFA LM3)

Corporate governance is the system by which companies are directed and controlled. The board is the primary mechanism protecting shareholders' interests.

Board Composition Requirements

Mix of executive (management) and non-executive (independent) directors.
Independent directors: no material business relationship with company.
Key committees: Audit, Remuneration (Compensation), Nomination.

CFA LM3: best practice = majority independent directors. Chair should be independent (not the CEO). Separate Chair/CEO roles prevent concentration of power.

Board's Core Responsibilities

Hire, monitor and replace CEO and senior management.
Approve strategy, major capital allocation decisions, M&A.
Oversee risk management and internal controls.
Set and review executive compensation.
Ensure accurate and transparent financial reporting.
Protect shareholder rights and engage with institutional investors.

Governance Failure Example: Enron (2001)

Enron's board approved related-party transactions that enriched executives at shareholder expense. Audit committee failed to challenge aggressive accounting (mark-to-market on speculative energy contracts). Auditor (Arthur Andersen) had conflicts of interest via consulting revenue.
Result: USD 70bn shareholder loss. Led directly to Sarbanes-Oxley Act (2002): mandatory CEO/CFO certification of accounts, independent audit committees, criminal penalties.

Governance Mechanisms — Internal and External Controls

CFA LM3 distinguishes internal governance (structures within the firm) from external governance (market forces and regulation outside the firm).

Mechanism	Type	How It Works
Board of directors	Internal	Monitors management, approves strategy, sets exec comp. Independent directors reduce agency risk.
Executive compensation	Internal	Align manager incentives with shareholder returns via stock options, restricted stock, clawbacks.
Independent audit	Internal/External	External auditor verifies financial statements. Audit committee oversees independence.
Shareholder meetings / proxy	Internal	Annual vote on board, resolutions, major transactions. Activist shareholders can agitate for change.
Takeover market / M&A	External	Poorly managed companies with depressed valuations become acquisition targets. Disciplines management.
Legal & regulatory framework	External	Company law, securities regulation (SEC, FCA), stock exchange listing rules. Mandatory disclosures.
Debt covenants	External	Bondholders impose restrictions on leverage, capex, distributions. Limit manager risk-taking at creditor expense.

Executive Compensation — Aligning Agent with Principal

Good compensation design: base salary + annual bonus (short-term) + long-term equity incentive (LTI). Vesting + clawback provisions.

CFA LM3: executive compensation should balance short-term performance with long-term value creation. Misaligned comp is a root cause of agency problems.

Component	Objective	Potential Misalignment	Best Practice
Base salary	Attract and retain talent	Too high base = no performance incentive	Benchmark to peers; review regularly
Annual cash bonus	Reward short-term performance	Encourages short-termism, earnings management	Cap bonus; tie to quality metrics; claw back if restated
Stock options	Align with shareholder returns	Options reward volatility, not just value creation. Out-of-money options worthless.	Restricted stock preferred: downside as well as upside
Restricted stock units (RSU)	Long-term retention + alignment	Value drops with market, not just company performance	Multi-year vesting (3-5 years). Performance conditions.
Clawback provisions	Recover comp if results misrepresented	Rare in practice. Legal enforcement complex.	Mandatory under SOX for certain violations. Expand scope.

SECTION 04

Capital Allocation (CFA LM5)

Types of Capital Investments — CFA LM5 Exhibit 2

CFA LM5 classifies all capital investments into four categories based on purpose and risk level. The type of project determines the risk-adjusted discount rate.

Maintain Business — Going Concern

Replace outdated facilities. Improve efficiency. Manage risk.
Low to moderate risk — within existing operations.
Example: upgrade ERP system, replace ageing machinery.
Discount at standard WACC.

Regulatory / Compliance

Imposed by laws, regulators or safety standards.
Not optional — must be done regardless of NPV.
Example: safety upgrades, environmental remediation, data protection compliance.
NPV may be negative but project is mandatory.

Expand Existing Business

Grow capacity or scope within known business lines.
Moderate risk — market is familiar.
Example: open new factory, expand retail footprint, enter

New Lines of Business

R&D, acquisitions in new sectors, new products.
High risk and uncertainty.

Capital Allocation Process: 3 Steps (CFA LM5)

Step 1: Idea generation. Step 2: Investment analysis — NPV/IRR/ROIC. Step 3: Planning and prioritisation. Ongoing: monitoring and post-investment review.

Net Present Value — CFA LM5 Gerhardt Corporation (Example 4)

$$NPV = \sum CF_t / (1+r)^t \quad \text{Accept if } NPV \geq 0 \text{ (increases shareholder wealth)}$$

Directly from CFA Vol 3, LM5, Example 4. Standard conventional cash flow pattern — outflow today, inflows over 5 years.

CFA LM5 GERHARDT CORPORATION — EXAMPLE 4

Investment: €50 million today. Expected after-tax cash flows: €16 million per year for Years 1–4, plus €20 million in Year 5. Required rate of return: 10%.

$$NPV = -50 + 16/(1.10)^1 + 16/(1.10)^2 + 16/(1.10)^3 + 16/(1.10)^4 + 20/(1.10)^5$$
$$= -50 + 14.545 + 13.223 + 12.021 + 10.928 + 12.418$$
$$= -50 + 63.136$$
$$= €+13.136 \text{ million}$$

Decision: ACCEPT. NPV ≥ 0. The investment increases shareholder wealth by €13.136m.
Interpretation: the firm exchanges €50m today for an investment worth €63.136m. Net wealth gain = €13.136m.

Excel: =NPV(0.10, 16, 16, 16, 16, 20) − 50 = 13.136

Unconventional Cash Flows & Multiple IRRs — CFA LM5

When cash flow signs change more than once, multiple IRRs may exist. CFA LM5: in this case, NPV should ALWAYS be used — IRR is unreliable.

CFA LM5 GERHARDT EXAMPLE 5 — Unconventional CFs (NPV = €4.78m)

CFs (in €m): t=0: −50 | t=1: +10 | t=1.5: −5 | t=2: +13 | t=3: +16 | t=4: +19 | t=5: +23
Note: one NEGATIVE cash flow mid-stream (t=1.5). Must use XNPV (not NPV) function.

NPV (at 10%) = €4.78 million. Since NPV ≥ 0, ACCEPT.

CFA LM5 — Multiple IRRs: When IRR Fails

If cash flows change sign more than once, the IRR equation can have MULTIPLE solutions.

Example (CFA LM5): t=0: −1,000 | t=1: +5,000 | t=2: −6,000

This has TWO IRRs: 100% and 200%. Both mathematically satisfy the IRR equation = 0.
Question: which IRR is correct? NEITHER tells you whether to accept or reject.

CFA RULE: when unconventional cash flows create multiple IRRs, use NPV.
Also applies when IRR and NPV conflict on MUTUALLY EXCLUSIVE projects — always follow NPV.

Internal Rate of Return — CFA LM5 Gerhardt (Examples 6 & 7)

IRR = discount rate that makes NPV = 0 Accept if IRR ≥ hurdle rate (required return)

Directly from CFA Vol 3, LM5, Examples 6 and 7. Two different projects with SAME initial investment but different outcomes.

CFA LM5 Example 6 — ACCEPT

Project: −50, +16, +16, +16, +16, +20 (same as NPV example).
IRR = 19.52%.

Decision: 19.52% > 10% hurdle rate → ACCEPT.
NPV (€13.136m) and IRR (19.52%) both agree → accept.

Excel: =IRR(−50, 16, 16, 16, 16, 20) = 19.52%

CFA LM5 Example 7 — REJECT

Project: −50, 0, 1, 2.5, 3, 16, 20, 25 (slower start, uneven timing)
IRR = 6.76% (using XIRR function for uneven periods).

Decision: 6.76% < 10% hurdle rate → REJECT.
NPV < 0 (consistent with IRR rule here).

Lesson: same initial investment, very different IRR.

Key limitation of IRR: assumes interim cash flows reinvested at IRR. If actual reinvestment rate < IRR, true return < IRR. NPV uses r (more realistic).

Return on Invested Capital — CFA LM5 Gerhardt (Example 8)

ROIC = After-tax operating profit / Average invested capital

Compare to WACC: ROIC > WACC = value creating

ROIC is what analysts can compute from public filings. Unlike NPV/IRR, ROIC is an aggregate measure across ALL investments (not project-by-project).

CFA LM5 EXAMPLE 8 — ROIC CALCULATION

Company reports Year 2 after-tax operating profit = 24,395.
Invested capital (long-term liabilities + equity):
End Year 1 = 112,257 + 15,688 + 148,442 = 276,387
End Year 2 = 106,597 + 15,688 + 159,995 = 282,280
Average invested capital = (276,387 + 282,280) / 2 = 279,333

ROIC = 24,395 / 279,333 = 8.73%

Q: If investor requires 10% return, is this company a promising investment?
A (CFA): NO. Although the company is profitable, ROIC (8.73%) < required return (10%). Investor would be better off elsewhere. The company is DESTROYING value relative to opportunity cost.

ROIC decomposition: ROIC = After-tax margin × Capital turnover. Two routes to higher ROIC: higher margins or higher asset efficiency.

NPV vs IRR — Conflict Resolution (CFA LM5)

When NPV and IRR conflict for MUTUALLY EXCLUSIVE projects: ALWAYS follow NPV

CFA LM5: NPV shows the actual increase in shareholder wealth in currency terms. IRR only shows a rate of return — it ignores scale and duration.

Project	Initial Investment	NPV at 7.5%	IRR	Decision (Choose One)
Project A	$33m	Negative NPV (−0.38m)	7.04% < 7.5% hurdle	REJECT (both NPV and I
Project B	$33m	Positive NPV (+$1.12m)	8.97% > 7.5% hurdle	ACCEPT (both agree)
Mutually Exclusive: H vs L	Different scales	H: higher NPV = $4m	L: higher IRR = 25%	Choose H: higher NPV a

CFA LM5 Practice Q4 — Project B: CF: 24, 16, 24, 7 at 7.5% hurdle

NPV = −33 + 0/(1.075) + 16/(1.075)² + 24/(1.075)³ + 7/(1.075)⁴ = $5.41m

IRR = 8.97% (CFA answer: accept because IRR > 7.5% hurdle rate)

Key CFA rule: when NPV and IRR conflict on mutually exclusive projects, choose HIGHER NPV project. IRR ignores project size. A

Cost of Equity — CAPM Approach (CFA LM8 Equity / LM6 Corp Finance)

$$r_e = r_f + \beta \times (r_m - r_f) \quad \text{[risk-free + equity risk premium × beta]}$$

The cost of equity is the return equity investors require. Unlike cost of debt (observable), equity cost is estimated — most commonly via the Capital Asset Pricing Model (CAPM).

Component	Definition	Typical Range	Comment
r_f (Risk-free rate)	Return on risk-free asset (gov bond)	3–5% in normal times	Use long-dated bond matching investment horizon
β (Beta)	Sensitivity of stock to market moves	0.5x (low) to 2.5x+ (high)	Estimated from historical regression vs market index
$(r_m - r_f)$ Market risk premium	Excess return of equity over risk-free	4–6% historically in developed markets	Long-run historical average or survey-based estimate
r_e (Cost of equity)	Minimum return equity holders require	6–15% typical range	Inputs must be forward-looking, not mechanical

Example: r_f=4%, β=1.2, ERP=5.5%

r_e = 4% + 1.2 × 5.5% = 4% + 6.6% = 10.6%. This stock requires 10.6% return.
A defensive utility (β=0.5): r_e = 4% + 0.5 × 5.5% = 6.75%. A volatile biotech (β=1.8): r_e = 4% + 1.8 × 5.5% = 13.9%.

Dividend Policy & Capital Returns (CFA LM6 Context)

Once capital allocation is complete and WACC is set, the firm must decide how to return excess capital. Three primary mechanisms.

Mechanism	How It Works	Tax Impact	Signal to Market	Best For
Regular cash dividend	Fixed per-share payment each period	Taxed as income in most jurisdictions	Commitment signal: hard to cut without negative impact	Mature, stable cash businesses
Special / extra dividend	One-off large distribution from windfall	Taxed as income	No commitment. One-time event.	After asset sale, lega settlement win
Share buyback (repurchase)	Company buys own shares in open market or tender	Often taxed at lower capital gains rate	Management confident shares are undervalued	Tax-efficient return. Flexibility to vary am

Dividend Irrelevance (Miller-Modigliani)

Under perfect markets, investors can manufacture their own dividend by selling shares. Policy doesn't affect value — only cash flow timing. Reality: taxes, signalling and clientele effects matter.

Pecking Order & Dividends

Firms set dividends at sustainable long-term earnings. Avoid cutting. Use excess FCF for buybacks (flexible). Dividends are sticky: a cut signals permanent earnings decline. Companies hate cutting.

SECTION 05

WACC & Cost of Capital (CFA LM6)

WACC — Formula & Nutry Company (CFA LM6)

$$WACC = w_9 \times k_9 \times (1-t) + w_e \times k_e \quad \text{[after-tax cost of debt + cost of equity]}$$

WACC is the required return for capital allocation decisions. It blends the cost of each financing source weighted by its proportion in the capital structure.

CFA LM6 — NUTRY COMPANY WACC = 8.2%

Nutry Company: 30% debt at 5%, 70% equity at 10%, corporate tax rate = 20%.

WACC = (0.30 × 5% × (1 − 0.20)) + (0.70 × 10%)
= (0.30 × 4.0%) + 7.0%
= 1.2% + 7.0%
= 8.2% (CFA answer B)

Why after-tax cost of debt? Interest is tax-deductible. Every £1 of interest reduces taxable income by £1, saving t on tax. Net cost = $k_9 \times (1-t)$.

Weighting approach: use MARKET value weights (not book). Market weights reflect current opportunity cost for investors.

Weight	How Determined	Preference
Market weights	Current market price × outstanding shares / bonds	Preferred: reflects actual investor opportunity costs
Book weights	Balance sheet carrying values	Less accurate: historical, not forward-looking

Leverage & Risk — Firm A vs Firm B (CFA LM6 Knowledge Check)

Directly from CFA Vol 3, LM6. Both firms have Assets=100, Revenue=100, Operating expenses=70. Firm A is conservatively financed, Firm B is highly levered.

Metric	Firm A (20% debt)	Firm B (60% debt)	−25% OI: Firm A	−25% OI: Firm B	+25% OI: Firm A	+ F
Operating Income	30	30	22.5	22.5	37.5	3
Interest expense	2	9	2	9	2	9
Profit	28	21	20.5	13.5	35.5	2
Total Equity	80	40	80	40	80	4
ROE	35%	53%	26%	34%	44%	7
Interest Coverage	15.0x	3.3x	11.3x	2.5x	18.8x	4

CFA Lesson

Leverage amplifies BOTH upside and downside. In the base case, Firm B has higher ROE (53% vs 35%), but when OI falls 25%, Firm B's coverage ratio drops to 2.5x (distress territory) while Firm A's remains comfortable at 11x. Firm A has greater stability and warrants lower costs of debt and equity.

Operating Leverage — Fixed Costs Amplify Business Risk

Operating leverage = proportion of FIXED costs in total costs. High fixed costs → higher earnings volatility.

CFA LM6: operating leverage is a key determinant of business risk. Firms with high operating leverage are riskier and warrant higher cost of capital.

Scenario	High Fixed Cost Firm	Low Fixed Cost Firm	Comment
Revenue falls 20%	EBIT falls 50%+	EBIT falls ~20%	Fixed costs don't move; hit EBIT hard
Revenue rises 20%	EBIT rises 50%+	EBIT rises ~20%	Operating leverage amplifies upside too
Breakeven	Higher: must cover large fixed base	Lower: covers as you go	Asset-heavy biz needs more volume to break even
Examples	Airlines, steel mills, chip fabs	Consulting, SaaS (variable server costs)	Capital-intensive = high op leverage
Capital structure impact	Lower financial leverage recommended	Can support more debt	Total leverage = Op leverage × Financial leverage

Degree of Operating Leverage (DOL) = % change in EBIT / % change in Revenue. DOL = Contribution Margin / EBIT. CFA LM6 Oldtech example: high fixed costs + revenue slowdown = disproportionate earnings decline.

Formula Reference Card — Session 7 Corporate Finance

All key formulas from this session.

NPV	$NPV = CF_0 + CF_1/(1+r) + CF_2/(1+r)^2 + \ldots + CF_t/(1+r)^t$ Accept if NPV ≥ 0
IRR	Rate r* where NPV = 0. Accept if IRR ≥ hurdle rate. Use NPV if multiple IRRs exist.
ROIC	ROIC = After-tax operating profit / Average invested capital
WACC	$WACC = w_9 \times k_9 \times (1-t) + w_e \times k_e$ Use market value weights
CAPM (cost of equity)	$r_e = r_f + \beta \times (r_m - r_f)$
MM Prop I (no taxes)	V_L = V_U (firm value independent of capital structure)
MM Prop II (no taxes)	$r_e = r_0 + (r_0 - r_9) \times (D/E)$ WACC unchanged
MM Prop I (with taxes)	V_L = V_U + tD (debt tax shield adds value)
MM Prop II (with taxes)	$r_e = r_0 + (r_0 - r_9)(1-t)(D/E)$ WACC falls with leverage
Static trade-off	V_L = V_U + tD − PV(Financial distress costs). Optimal at D*.
Degree of Operating Leverage	DOL = % ΔEBIT / % ΔRevenue = Contribution Margin / EBIT
Interest coverage	EBIT / Interest expense. < 1.5x = concern.
Net debt / EBITDA	(LT Debt + ST Debt − Cash) / EBITDA. < 2x low; > 5x high.

SECTION 06

Capital Structure Theory (CFA LM6)

Modigliani-Miller Propositions — Without Taxes (CFA LM6)

The MM framework starts with a simplifying assumption: no taxes, no bankruptcy costs. Under these conditions, capital structure is IRRELEVANT.

MM Proposition I (No Taxes)

$V_L = V_U$
The value of a levered firm = value of unlevered firm.
Firm value is determined SOLELY by future cash flows, not how they are financed.

Intuition: in perfect capital markets, investors can replicate any capital structure themselves (home-made leverage). The firm adds no value by choosing debt vs equity.

MM Proposition II (No Taxes)

$r_e = r_0 + (r_0 - r_d) \times (D/E)$

Levered equity cost RISES as leverage increases.
Lower-cost debt is offset EXACTLY by higher-cost equity.
Result: WACC is UNCHANGED regardless of capital structure.

Intuition: adding cheap debt increases equity risk, so equity investors demand more. The two effects cancel perfectly.

Gerhardt Under MM Prop I & II (No Taxes) — CFA LM6

Gerhardt is all-equity, WACC=10%, perpetual CF=€5,000. V=€5,000/0.10=€50,000. Issues €15,000 debt at 5%, buys back equity.
MM Prop I: V stays at €50,000 (D=–15k, E=–35k).
MM Prop II: new re = 10% + (10%–5%) × 15,000/35,000 = 12.143%. WACC = (15/50)(5%) + (35/50)(12.143%) = 10% — unchanged. ✓

MM With Taxes — CFA LM6 Plover Corp (Practice Q5)

Once we add corporate taxes, the analysis changes: interest is tax-deductible, so debt creates a tax shield that INCREASES firm value.

MM Prop I with taxes: $V_L = V_U + tD$ (debt tax shield increases firm value)

CFA LM6 PRACTICE Q5 — PLOVER INC.

Plover: unlevered firm, V_U = €100m, perpetual pre-tax CF = €10m/yr, t = 20%, r_0 = 8% (WACC = CF(1−t)/V = 8÷100 = 8.0%).
Plans to issue €35m debt at r_d = 4.5% (financial distress costs ≈ zero).

Step 1: $V_L = V_U + tD$ = €100m + 0.20 × €35m = €107m
Step 2: E = V_L − D = €107m − €35m = €72m
Step 3: re = $r_0 + (r_0 - r_d)(1-t)(D/E)$ = 8% + (8%−4.5%)(0.80)(€35m/€72m) = 9.36%
Step 4: WACC = (35/107)(4.5%)(0.80) + (72/107)(9.36%) = 1.18% + 6.30% = 7.48% (CFA answer B)

Key: WACC fell from 8.0% (unlevered) to 7.48% (levered). Debt tax shield reduces WACC and increases firm value.

Static Trade-Off Theory — The Optimal Capital Structure (CFA LM6)

MM with taxes (no distress) implies 100% debt is optimal. Reality: financial distress costs grow non-linearly with debt. The static trade-off theory finds the optimal balance.

$$V_L = V_U + tD - PV(\text{Financial Distress Costs})$$

Optimal D* maximises V_L

The Trade-Off

As debt increases: (1) Tax shield rises (value +tD). (2) Financial distress probability increases (value −PV distress).

At LOW debt: tax shield > distress cost → V_L rises. Add more debt.
At D*: marginal tax benefit = marginal distress cost. This is the OPTIMAL capital structure.
Beyond D*: distress cost > tax benefit → V_L falls. Reduce debt.

CFA Boulder example: V_U=$45m, t=30%. Optimal D=$20m (where V_L=$49.5m is maximised). At D=$30m, distress cost swamps tax shield and V_L falls to $42m.

Financial Distress Costs

Direct: legal fees, restructuring advisors, management time in bankruptcy.
Indirect: customers leave (fear of bankruptcy), suppliers tighten terms, top talent exits, underinvestment as equity views

Real-World Implication

Firms do NOT actually reach 100% debt. Static trade-off explains why optimal leverage differs by industry: stable, tangible-asset companies (utilities, real estate) carry more debt than high-

Pecking Order Theory & Real Options (CFA LM6 / LM5)

The pecking order theory explains observed financing behaviour through information asymmetry. Real options add optionality value that NPV misses.

Pecking Order Theory (Myers-Majluf 1984)

Managers prefer financing with least information content:
1st: Internal funds (retained earnings) — no signal, no flotation cost.
2nd: Debt — signals confidence (commitment to fixed payments).
Last: Equity issuance — often seen as a NEGATIVE signal.

Reasoning: if managers issue equity, market infers shares are overvalued. Equity issuance → price drop.
No optimal capital structure: firms just follow the hierarchy.

Real Options

Real options = right (not obligation) to take a future action that changes investment value.

Types:
- Option to EXPAND: if demand is strong, invest more (e.g. build second factory)
- Option to ABANDON: exit if project underperforms (limits downside)
- Option to DELAY: wait for more information before committing
- Flexibility option: change product mix

Value = NPV + Real Option Value

CFA LM5 on Real Options

Real options are similar to financial options but deal with real assets. They grant companies the right to make a decision in the future that alters the value of capital investments made today. By providing future decision-making flexibility, real options can be an important piece of the value in many capital investments.

PRACTICE PROBLEMS

Work through each before checking answers.

Q1 Gerhardt Corp (CFA LM5 Ex4): Investment=€50m today. CFs: €16m/yr for Years 1-4, plus €20m in Year 5. Required return=10%. Calculate NPV. Should Gerhardt invest?

Q2 Gerhardt IRR (CFA LM5 Ex6): Same CFs as above: −50, +16, +16, +16, +16, +20. IRR=19.52%. If hurdle rate is 10%, should Gerhardt accept? What if hurdle rate is 20%?

Q3 Nutry WACC (CFA LM6): 30% debt at 5%, 70% equity at 10%, tax=20%. Calculate WACC. Identify which answer is correct: A. 5.9% B. 8.2% C. 8.5%.

Q4 Plover Corp (CFA LM6 Q5): V_U=€100m, t=20%, D=€35m, rd=4.5%, r_0=8%. (a) Calculate V_L. (b) Calculate equity value E. (c) Calculate re. (d) Calculate WACC.

Q5 ROIC (CFA LM5 Ex8): After-tax operating profit=24,395. Average invested capital=279,333. Calculate ROIC. If required return=10%, is this company a promising investment?

Practice — Worked Solutions

A1

NPV = −50 + 16/1.10 + 16/1.10^2 + 16/1.10^3 + 16/1.10^4 + 20/1.10^5 = −50 + 14.545+13.223+12.021+10.928+12.418 = −50 + 63.136 = +€13.136m. Decision: ACCEPT (NPV ≥ 0). Company increases shareholder wealth by €13.136m. Excel: =NPV(0.10,16,16,16,16,20)−50 = 13.136.

A2

IRR=19.52% > hurdle 10% → ACCEPT. If hurdle = 20%: IRR=19.52% < 20% → REJECT. The IRR tells us the rate of return earned by the project. It only exceeds the required return when return is < 19.52%. NPV and IRR agree here (both conventional CFs, single sign change).

A3

WACC = (0.30 × 5% × 0.80) + (0.70 × 10%) = (0.30 × 4%) + 7% = 1.2% + 7.0% = 8.2%. Answer B. Not A (5.9% ignores equity). Not C (8.5% uses pre-tax debt cost of 5% not after-tax 4%).

A4

(a) V_L = V_U + tD = 100 + 0.20 × 35 = €107m. (b) E = V_L − D = 107 − 35 = €72m. (c) re = 8% + (8%−4.5%)(1−0.20)(35/72) = 8% + 3.5% × 0.80 × 0.486 = 8% + 1.36% = 9.36%. (d) WACC = (35/107)(4.5%)(0.80) + (72/107)(9.36%) = 1.18% + 6.30% = 7.48%. CFA answer B confirmed.

A5

ROIC = 24,395 / 279,333 = 8.73%. Since ROIC (8.73%) < required return (10%), the company is NOT a promising investment. Despite being profitable, it generates less return than investors could earn elsewhere on a similar-risk investment. Management should improve margins or asset turnover, or return capital.

KEY TAKEAWAYS

Session 7 — Corporate Finance

- Gerhardt NPV (CFA LM5): −50+PV(16 annuity)+PV(20)=+€13.136m. Accept. NPV Ex2=€4.78m (unconventional CFs).
- Gerhardt IRR (CFA LM5): 19.52%>10% → accept (Ex6). 6.76%<10% → reject (Ex7). Multiple IRRs (−1000,+5000,−6000) → use NPV.
- ROIC (CFA LM5): 24,395/279,333=8.73%<10% required → not a promising investment (value destroying vs opportunity cost).
- Nutry WACC (CFA LM6): (0.3)(5%)(0.8)+(0.70)(10%)=8.2%. Cost of debt is after-tax. Always use market value weights.
- Firm A (20% debt): ROE=35%, coverage=15x. Firm B (60% debt): ROE=53%, coverage=3.3x. OI−25%: Firm B coverage=2.5x.
- Plover (CFA LM6): V_L=€107m=100+0.2×35. E=€72m. re=9.36%. WACC=7.48% (down from 8.0% unlevered).

- MM: no taxes → V_L=V_U, WACC unchanged. With taxes →
- Static trade-off: optimal D* where marginal tax